

SPARKID

SparkID generates broker authorization from a declared statement of intent, so a Sparkplug estate's access rules are never hand-typed wildcards.

— VERSION 0.1.0

IGNITION 8.1.19+ AND 8.3

RUNS ON: HOST

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1. OVERVIEW

SparkID generates broker authorization for a Sparkplug estate from a declared statement of intent, so a wildcard authorization rule is never hand-typed. An estate needs one identity per edge node, host, replay writer and consumer, each with authorization rules written against the Sparkplug topic grammar; SparkID reads a `.intent` declaration, expands it against that grammar per message type and scope, and emits the broker artifacts. It never writes into a running broker and never opens a network connection; installing the generated artifacts and restarting the broker is a separate, deliberate step.

- A SparkBridge Host gateway.
- Java 17 if building or running the CLI from source.
- A supported broker target: HiveMQ file-RBAC (`credentials.xml`) or Mosquitto dynamic-security (`--mosquitto`, writes `dynamic-security.json`).

3. INSTALL

SparkID comes with the Host license. Install the module through Config > Modules > Install or Upgrade a Module; no build step is required for the packaged release.

To build the CLI from source:

```
JAVA_HOME=~/.local/opt/jdk-17.0.19+10/Contents/Home ./gradlew fatJar  
java -jar build/libs/sparkid-0.1.0-all.jar --help
```

The jar depends on `io.sparkbridge:common` from `mavenLocal`; publish it first if missing:

```
cd ../sparkbridge && ./gradlew :common:publishToMavenLocal
```

Soft-licensed; SparkID runs the same whether or not a key is present. License file lookup order: `--license <file>`, the `SPARKBRIDGE_LICENSE` environment variable, then `./sparkbridge-license.key` in the working directory. Run `sparkid license` for full status.

5. CONFIGURE

The `.intent` file is a declarative access policy meant for infrastructure-as-code review and diffing. The workflow, start to finish:

1. Validate the written declaration:

```
sparkid validate --intent examples/sparkbridge-dev.intent
```

2. Issue one secret per identity and record it in the ledger:

```
sparkid issue --intent examples/sparkbridge-dev.intent --ledger out/estate.ledger
```

3. Preview the change against the broker's current rules:

```
sparkid diff --intent examples/sparkbridge-dev.intent --current /path/to/credentials.xml
```

4. Generate the reviewed deployment artifacts:

```
sparkid generate --intent examples/sparkbridge-dev.intent --ledger out/estate.ledger
```

BROKER TARGETS

HiveMQ file-RBAC is the only supported target for `credentials.xml`. Mosquitto dynamic-security is supported with `--mosquitto`, writing `dynamic-security.json`.

WRITE BEHAVIOR

SparkID never writes into a running broker and never opens a network connection; installing generated artifacts and restarting the broker is a separate, deliberate step.

SparkID is a CLI workflow, not a running service: validate the intent, issue secrets, preview the diff, generate artifacts, then deploy the generated files to the broker by hand and restart it. Re-run `sparkid issue` with the same ledger to rotate credentials inside the overlap window the rotation logic supports.

7. VERIFY

Run `sparkid diff` against the broker's current `credentials.xml` and confirm the reported changes match intent. Check credential age with `sparkid expiry`. Full acceptance runs are documented in `docs/ACCEPTANCE.md` in the repo.

SYMPTOM	CAUSE	FIX
	The declared intent would lock out the primary host identity	Review the intent; SparkID refuses a generated policy that would cause host lockout
	Generated artifacts were not installed and the broker was not restarted	Copy the generated <code>credentials.xml</code> or <code>dynamic-security.json</code> into place and restart the broker; SparkID never does this for you

SYMPTOM

CAUSE

FIX

Not published
to mavenLocal

Run `./gradlew
:common:publishToMavenLocal` in
the SparkBridge repo

9. REFERENCE

- CLI commands: `validate`, `issue`, `diff`, `generate`, `expiry`, `license`.
- Broker targets: HiveMQ file-RBAC (default), Mosquitto dynamic-security (`--mosquitto`).
- License file: `sparkbridge-license.key` (working directory, `--license`, or `SPARKBRIDGE_LICENSE`).

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